REFERENCES

Used standard open-source libraries like Whisper, Librosa, and PyTorch, but the pipeline integration, LID model, and processing logic were implemented independently.

Libraries / Tools Used

1. OpenAI Whisper
   Used for speech-to-text transcription of code-switched audio.
2. Librosa
   Used for MFCC extraction, pitch (F0), and energy computation.
3. PyTorch
   Used to build and train the language identification (LID) model.
4. SpeechBrain
   Used for speaker embedding extraction.
5. gTTS
   Used for generating synthetic Tamil speech.
6. eng_to_ipa
   Used for converting English text into IPA representation.
7. Scikit-learn
   Used for implementing the spoof detection classifier.

Concepts / Methods Referenced

8. Mel-Frequency Cepstral Coefficients (MFCC)
   Standard feature extraction technique for speech processing.
9. Dynamic Time Warping (DTW)
   Used for aligning temporal sequences of speech signals.
10. Logistic Regression
    Used for binary classification (real vs synthetic speech).

Datasets / Audio

11. You tube machine learning hinglish video
12. Student-recorded reference audio as student_voice_ref.wav